

Beyond the customer: Opening the agile systems development process

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ABSTRACT

Context: A particular strength of agile systems development approaches is that they encourage a move away from ‘introverted’ development, involving the customer in all areas of development, leading to more innovative and hence more valuable information system. However, a move toward open innovation requires a focus that goes beyond a single customer representative, involving a broader range of stakeholders, both inside and outside the organisation in a continuous, systematic way.

Objective: This paper provides an in-depth discussion of the applicability and implications of open innovation in an agile environment.

Method: We draw on two illustrative cases from industry.

Results: We highlight some distinct problems that arose when two project teams tried to combine agile and open innovation principles. For example, openness is often compromised by a perceived competitive element and lack of transparency between business units. In addition, minimal documentation often reduce effective knowledge transfer while the use of short iterations, stand-up meetings and presence of on-site customer reduce the amount of time for sharing ideas outside the team.

Conclusion: A clear understanding of the inter- and intra-organisational applicability and implications of open innovation in agile systems development is required to address key challenges for research and practice.

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1. Introduction

The last 10 years or so has seen the emergence of a number of agile systems development (ASD) methods, such as XP [5] and Scrum [53]. These methods have been well received by those in ISD and there is strong anecdotal evidence to suggest that awareness and indeed use of these methods is highly prevalent across the community. Agile has been described as ‘the business of innovation’ [38], relying on people and their creativity rather than on processes [16]. Additionally, Cockburn [15] contends that “agile approaches are best employed to explore new ground and to power teams for which innovation and creativity are paramount”. Agile methods, given their flexible and light-weight processes, place emphasis on close communication and collaboration in project teams [6,53]. However, some reports have heavily criticised what agile research exists (e.g. [1,20,26]). These reports accuse the current body of agile method research of lacking rigor, cumulative tradition and sufficient theoretical grounding. They even point to the ambiguity as to what constitutes ‘agility’, stating that it “now means so many things to so many people, it has lost a lot of its meaning” [20].

A particular strength of agile approaches is that they move away from ‘introverted’ development where the team building the system is detached from the customer. Instead, agile approaches continually involve the customer in the development process, supposedly leading to the development of a more innovative and hence more valuable information system [5,53]. Despite these claims, however, there is a lack of understanding as to the actual extent that agile methods facilitate such innovation. Even if we had reliable evidence of the current state of agile innovation, we believe a wider interpretation of agile needs to be adopted. While the customer plays an essential part in the agile process, this practice could be extended to include multiple stakeholders and even other organisations. We propose that it is useful to consider how the agile innovation process can benefit from becoming more ‘open’, e.g., by opening up the boundaries of a systems development entity to include other stakeholders besides that of the customer. For example, it has been suggested that companies must increasingly work with each other to enhance their agility in adapting to market developments and developing new products/services cheaper and faster [57].

As far as we are aware, no research has focused on the role of other stakeholders in agile development besides the customer. Nor has research looked at how principles of open innovation could complement an agile approach, despite the commonalities between the two models, particularly its

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emphasis on the value of people and communications. Thus, exploring the notion of open innovation and its applicability and implications in a multiple project environment that employ agile methods is timely. This paper builds on an earlier report of the study [19]). In this version the paper provides a more in-depth discussion of the motivations behind examining open innovation in an agile context. It also provides a more detailed discussion of the open innovation literature as well as providing illustrative empirical cases to supplement the argument. The remainder of the paper is structured as follows. The next section presents the current state of research on innovation in agile development, highlighting some key shortcomings, most critical of which is a lack of rigorous empirical research on the extent to which agile approaches facilitate and drive innovation. Next, some contemporary views on innovation are described, most notably the principles of open innovation. An open innovation conceptual framework is put forward, with the intention of driving future research in this area. Drawing on two illustrative case studies, we highlight some distinct problems that arose when these two project teams tried to combine agile and open innovation principles. Some avenues for future research are then put forward, followed by a discussion of the limitations of the study.

2. Innovation in agile development – the current state of research

Innovation and creativity have been advocated as a core part of Information Systems Development (ISD) for many years [9,11,22,27,34,44,52]. According to Sternberg and Lubart [56], creativity is the ability to produce work that is considered novel, appropriate and adaptive. Indeed Cougar [22] believed that creative and innovative activities should play a pivotal role “in all aspects of IT development, from requirements definition through program design”. Three reasons have been proposed for this. Firstly, “technology is evolving on a daily basis and we can continually look for new ways to utilise resources”. Secondly, “most simple systems have already been developed and the challenging ones are still ahead”. Finally, “many information systems are old, not meeting existing demand, and will soon become obsolete” [44]. In addition, researchers such as Gallivan [34] highlight the importance of creative developers, and Brooks [9] even contends that the critical problems in ISD may not be addressed by ISD methods *per se*, but rather by how those methods facilitate creativity and improvisation.

The importance of creativity has also been highlighted and the support towards creativity claimed within the agile method movement [15,16,38]. Agile advocates believe that “creativity, not voluminous written rules, is the only way to manage complex software development problems” [38]. Cockburn and Highsmith [16] also claim that “agile methodologies deal with unpredictability by relying on people and their creativity rather than on processes”. Additionally, it has been contended that “agile approaches are best employed to explore new ground and to power teams for which innovation and creativity are paramount” [15]. The literature also illustrates the fact that the requirement for creativity has been highlighted in discussions of specific agile methods, such as eXtreme Programming (XP), one of the most popular agile methods [8,15,23]. Additionally, Cockburn [15] observes that “although XP contains certain disciplined practices, its intent is to foster creativity and communication”. Indeed, Benediktsson et al. [8] claim that “given the benefits of XP in terms of creativity, value delivery and higher satisfaction levels, it is not surprising that many managers and developers have adopted such practices”.

Despite these claims, however, there is a lack of understanding of what constitutes innovation in software development in general and to what extent agile methods actually facilitate this process. Given that agile approaches are rarely, if ever, implemented in their textbook, ‘vanilla’ format [18], it is difficult to claim that these aspirational claims to increase innovation and creativity are actually occurring in practice. Some more rigorous studies have examined the relationship between agility and innovation or improvisation [63,65,66], but these have tended to focus more on the agile practices themselves as the innovation and not the extent to which the practices facilitate agility and innovation.

3. Contemporary thinking on innovation – open innovation

Innovation is now viewed as the lifeblood of organizations that want to survive and prosper in a marketplace that is global in nature and intensely competitive. However, this particular stance on the importance of innovation did not always exist. Traditionally, the innovation process has taken a linear approach, the expectation being that investment in research and development would provide organisations with a competitive advantage [39]. In addition, conventional approaches to innovation assumed that it was the experts ‘within’ the company that invented and designed innovative new products to meet customer needs and organisations rarely looked outside for new ideas or inventions [57]. As Prahalad and Hamel [49] pointed out, “organizations often tend to be hidebound and so orthodox ridden, “that the only way to innovate is to put a few bright people in a dark room, poor in some money, and hope that something wonderful will happen” (p. 66). This ‘Silicon Valley’ approach resulted in innovation being an isolated activity where growth depended on the inventive capacity of individuals and small teams [49]. Thus, this very approach in which organisations generate, develop and commercialise their own ideas belong to the closed model of innovation [30].

Closed innovation is a view that successful innovation requires control and that firms need to be strongly self-reliant because of uncertainty with quality, availability and capability of others’ ideas [12]. Traditionally, new business development processes and the marketing of new products took place within the firm boundaries and exclusively with internal resources. Within the closed model, the innovation process is characterised by firms that invest in their own R&D, employing smart and talented people in order to outperform their competitors in new product and service development. In addition, after producing a stream of new ideas and inventions, firms must defend their intellectual property thoroughly against the competition [24]. Changes in society and industry, however, have led to an increased availability and mobility of knowledge workers and the development of new financial structures like venture capitalism. Indeed, Gassmann and Enkel [35] propose that shorter innovation cycles, industrial research and the rising costs of development, in addition to a lack of resources are motives that are changing companies’ innovation strategies towards a more open direction [35].

Thus, it has been argued that a paradigm shift is taking place in how companies commercialise knowledge, resulting in the boundaries of a firm eroding. According to Nonaka and Toyama [47] and Tidd et al. [59], successful innovation is the result of combining different knowledge sets and such knowledge is frequently to be found outside the organisation [12,25]. For example, empirical evidence from industry studies carried out by Patel and Pavitt [48], Prencipe [50] and Brusoni et al. [10] suggest that a firm’s knowledge boundaries extend far beyond their production boundaries. Indeed, Cohen and Levinthal [17](p. 133)

claim that “interactions among individuals who each possess diverse and different knowledge structures will augment the organisation’s capacity of making novel linkages and associations... innovating beyond what any individual can create alone”. In addition, it has been found that large amounts of knowledge are a necessary condition for creativity to occur in firms, which in turn translates into innovative ideas [4,56]. Open Innovation has emerged as a model where firms commercialise both external and internal ideas that can be taken to market through external channels, outside a firm’s current business, to generate value [12]). This paradigm is viewed as the extreme opposite of the traditional model of innovation where research and development activities lead to internally developed products that were then distributed by the firm [14]. A general theme underlying open innovation is that firms cannot continue to look inward in their innovation processes, isolating themselves from possible partners, collaborators and competitors. In other words, open innovation invites firms to open up their boundaries to achieve a flexible and agile environment. The term ‘open innovation’ has been defined by West and Gallagher [68](p.82) “as systematically encouraging and exploring a wide range of internal and external sources for innovation opportunities, consciously integrating that exploration with firms capabilities and resources and broadly exploiting those opportunities through multiple channels”. In addition, Laursen and Salter [42] focused on external search breadth and external search depth for different types of innovation in a large-scale sample of UK manufacturing firms. These authors defined openness as “the number of different sources of external knowledge that each firm draws upon in its innovative activities” (2004, p.1204).

In contrast to the linear closed model of innovation, the open innovation approach suggests that firms develop processes to ensure a flow of ideas across its boundaries because not all smart people work for the organisation and there is an increasing geographical dispersion of knowledge [24]. Thus, ideal business search outside their own companies for the best ideas, seeking input from other companies, which include competitors, as well as from customers, suppliers and vendors. However, the concept of open innovation challenges existing theories of innovation adoption due to changes in organisational control (cf. [12] and risk (cf. [13]. Indeed it has been argued that open innovation is far more complicated than it seems because it is not always that straightforward for organisations to have a high degree of openness and it can also be costly to implement. Thus, it is necessary to identify the challenges of an open innovation approach in order to provide credible insights for practitioners [24]. In this regard, exploring the applicability and implications of open innovation principles in an agile environment is a step in the right direction.

4. A conceptual framework to guide research

For our theoretical base, we propose a framework drawn from three central open innovation archetypes proposed by Gassmann and Enkel [35]. These include: (1) the outside-in process; (2) the inside-out process; and (3) the coupled process. We feel this framework provides a useful lens to examine the applicability and implications of open innovation in an agile environment. For example, open innovation can be analysed at a number of levels, which include the intra-organisational and inter-organisational networking level [14]. However, while most existing research focuses on inter-organisational aspects, open innovation also increases the significance of intra-organisational networks [69]. Indeed, the effective management of externally acquired knowledge will most likely require the development of complementary

internal networks [37,69]. The implications, however, that open innovation has *within* an organisation and in particular the fact that it affects different parts of an organisation differently are largely neglected in the current literature [2]. In other words, while there exists much research about intra-organisational level networking in general to stimulate innovation (e.g., [32,40,60], this type of networking has not been analysed explicitly within the open innovation context [61]. Most importantly, there is no research that we know of that addresses either intra-organisational or inter-organisational networking in an agile environment. In order to tackle this gap, we have tailored Gassmann and Enkel’s framework to include innovation that occurs outside the boundaries of both a business unit, which addresses the level intra-organisational networking, and the firm, which addresses the inter-organisational level (see Fig. 1).

4.1. The outside-in process

Companies that decide on an outside-in process (also called inbound open innovation) as a core open innovation approach choose to cooperate with suppliers, customers third parties etc. and integrate the external knowledge gained. This can be achieved by investing in global knowledge creation, applying innovation across industries, customer and supplier integration, and purchasing intellectual property. Gassmann and Enkel [35] suggest that if firms possess the necessary competencies and capabilities, they can successfully integrate internal company resources with the critical resources of other members such as customers, suppliers etc., by extending new product development across organizational boundaries. For instance, in exploiting open source software as part of an open innovation approach, companies such as HP and Sun have used an outside-in process in the past by donating R&D to the Mozilla open source project while exploiting the pooled R&D and knowledge of all contributors (academics, user organizations, individual hobbyists) to facilitate the sale of related products. The result was that these firms maximised the returns from their innovation by concentrating on their own needs and then incorporating the shared browser technology into their own integrated systems. IBM is another notable example of a company that invests substantially in contact with customers and other external knowledge sources. The relevance of customer integration within a company’s innovation process has been illustrated in plenty of empirical studies (see for example, [51,55,64] and as already mentioned, agile methods actively involve the customer in the development process. However, it is worthwhile considering the involvement of other sources such as complimentary partners, suppliers, and network partners outside the boundaries of the firm, in addition to internal sources such as other business units within the organisation. We predict that a combination of different competencies and knowledge sets will enrich the agile innovation process.

Conversely, in order to apply the outside-in process, a company will require the absorptive capability to do so. Absorptive capacity, or the ability to understand, acquire, use and ultimately take advantage of knowledge inside and outside the firm has been long been recognised as being crucial for innovation [17,41,45]. As has also been pointed out by Cohen and Levinthal [17], “the ability of a firm to recognise the value of new, external information, assimilate it and apply it to commercial ends is critical to its innovative capabilities”. Absorptive capacity is even more important today due to the substantial increase in the external acquisition and external exploitation of knowledge assets by which companies attempt to complement their knowledge-bases. Thus, it is important to recognize that the efficiency of both knowledge generation and application in an agile environment is dependant on the concept of absorptive capacity. Additionally, it has been found that the

presence of an internal project champion that interacts effectively with others in many different functions across the organization is an important asset in the outside-in process [13].

Illustrative Case 1: ABC consulting¹ – difficulties with outside-in and agile.

Following expert advice and reading agile method literature one business unit in ABC Consulting decided to adopt XP and implemented 10 of the 12 XP practices on a project. One of these practices was the ‘on-site customer’, which they implemented rigidly as described in Beck [5]. However, the products being developed was for use by four other distinct business units within the organisation, and so an ‘on-site customer’ was assigned from each unit in the interests of generating and incorporating ideas from each of the business units. Each of the four representatives were invited to work with the team on a day-to-day basis and to all planning and retrospective meetings. While there were many positive aspects to this arrangement, there were many significant problems.

Firstly, while every representative participated intensively during the initial days and weeks of the project, this soon became more sporadic and disengaged in some cases. This resulted in inequity, according to one of these representatives, where the requirements and ideas getting supported were the ones associated with representatives that “had enough free time to be there”. According to another representative, this paradoxically resulted in the requirements of the least utilised business units being satisfied. Secondly, even when all representatives were present, some felt that the decisions regarding which requirements made it into the prioritised list at each iteration often based on politics, “crafty negotiation”, and “who could shout loudest and longest”, as opposed to any clear rationale or structured process. Thirdly, openness was compromised by a perceived competitive element between some units, where some representatives were very happy to discuss the ideas of others while being very protective of their own which may be valuable to the group. Finally, according to the manager, the quality of the ideas brought in was somewhat comprised as the representatives were relatively junior and had “extensive knowledge of their own day-to-day job but not of the wider organisation”.

IBM is yet again another example of a company that has successfully utilised an inside-out approach as part of its open source initiative that represented spinouts in the 1990s and more recently, donated software patents to the open source community [67].

Outsourcing has also been recognized as one mechanism that can also be used to channel knowledge and ideas to the external environment. The benefits of outsourcing include gaining access to new areas of complementary knowledge, managing capacity problems which allows for more flexibility, reduced time-to-market, sharing of costs and concentration of core competencies. It has been found that companies that choose an inside-out process are mainly research-driven companies like Pfizer or Novartis. Such companies aim to decrease their fixed costs of R&D and share the risks by outsourcing part of their development process [35]. However, it has been found that the inside-out process has not been adopted on a wide scale. Thus, a broader sample or examination of a more diverse range of organizations is necessary [13]. In the context of this study, an inside-out process refers to leveraging and transferring knowledge to stakeholders outside the boundaries of a business unit and firm, and gaining certain advantages by letting ideas flow to the outside.

Illustrative Case 2: WebCo² – difficulties with inside-out and agile.

A team in WebCo implemented a combination of XP and Scrum practices in early 2009, this team of nine staff operates in a large R&D group of 170 staff across approximately 15 teams. Another team were responsible for idea generation, looking across projects for ideas, and potential to exploit synergies. According to a senior developer on the team, ideas, experimentation and creativity increased dramatically with the transition to the agile approach. The transition “allowed the team to invent and develop” with “minimal distraction with documentation, useless team meetings and plans that never get followed”.

However, the new approach caused significant problems in terms of transferring the ideas outside the team. Firstly according to members of the over-arching team across all projects in the R&D group, “visibility as to what the new [agile] team were doing dropped very quickly” following the transition. On one hand there was the fact that the agile team were releasing software on a weekly basis thanks to the new process. In theory, however, that gave members of the other teams in R&D the opportunity to see exactly what was being produced. According to those interviewed, most did not take the opportunity to visit the evolving website, preferring the monthly review of documentation to identify synergies and new ideas. Because the agile approach focused on eliminating or at least minimising documentation, this project no longer featured heavily at these meetings. When probed as to why the working software was not utilised, the consensus seemed to be that, while it was significantly better for technically savvy staff, it did not suit members of other functions such

4.2. The inside-out process

This process (also referred to as outbound open innovation) focuses on the externalising of company knowledge and innovation in order to bring ideas to market faster. This approach includes licensing IP or multiplying technology by transferring ideas to other companies. Additionally, focusing on an inside-out process by commercialising ideas to different industries can increase a company’s revenue base substantially. For example, pharmaceutical companies like Novartis, Pfizer and Roche are well recognised for developing substances that were originally aimed at treating one ailment but became better known when used for others [35].

¹ This data was collected during a study in one multinational consulting organization in June 2009, using semi-structured, 1–2 h interviews with the lead partner, project managers and customer representatives. The interviews were recorded, transcribed and analysed using standard coding techniques, using codes based on the conceptual framework presented in Section 4. ‘ABC Consulting’ is a pseudonym used to protect anonymity.

² This data was collected in June 2009, using semi-structured, 1–2 h interviews with developers and the project managers. The interviews were recorded, transcribed and analysed using standard coding techniques, using codes based on the conceptual framework presented in Section 4. ‘WebCo’ is a pseudonym used to protect anonymity.

as marketing and accounting. In the absence of supporting explanatory documentation, most of this contingent did not log in to the system and try various functionality. Secondly, the introduction of agile coincided with a rapid drop in the number of developers from that team attending the overall R&D meetings. When asked about this drop, members of the team said that the tolerance for “prolonged” meetings dropped. According to one senior developer, “using the old plan-driven approach we would have been going to meeting after meeting, but since going to agile, every minute you spend in one of these meetings you just think about all of the work not being done”. Thirdly, some on the team felt that the opportunity for bringing ideas out to the R&D group level were limited by agile. The use of short iterations, a feature backlog and stand-up meetings reduced the amount of time “you can spend playing around or sharing ideas outside your team” (Senior Developer). The on-site customer practice was considered the most restrictive, as because you were accountable to a person there at all times, it was “harder to justify taking a half day out to sit with folks on other projects for the benefit of other customers.

4.3. The coupled process

This open innovation approach combines the outside-in (gaining external knowledge) with the inside-out process (to bring ideas to market). In order to accomplish both, these companies collaborate and cooperate with other stakeholders such as partner companies (e.g. strategic alliances, joint ventures), suppliers and customers, as well as universities and research institutes.

Alliances with complementary partners can lead to the occurrence of cooperative innovation processes. To collaborate and cooperate successfully, a give and take of knowledge approach is crucial. Benefits of such an approach include an intensive exchange of knowledge and a mutual learning process [35]. For the purpose of this research a couple process also refers to how business units

cooperate and interact with other business units in intra-organisational networks, in addition to inter-organisational networks. However, the notion of a so-called value network (a concept considered extremely important in the open innovation paradigm) should also be explored in future research. These networks include suppliers, partners, allies, competitors, customers and other network players working together. They focus on creating value, where value is determined by the resources and capabilities assembled and combined by different partners and how well they perform joint tasks [49,62]. Gaining entry to a network enables firms to address a specific knowledge need quickly, without having to spend an astronomical amount of time and money developing that knowledge internally or acquiring it through vertical integration [54]. In addition, cooperation between firms increases knowledge gain and reduces the waste of repeated effort [58]. Access to complementary skills and a broader knowledge-base that facilitates different types of knowledge exchange in a network positively influences firm innovation. Such knowledge is not limited to technical knowledge and may include knowledge from customers, market segments, partners, etc. that is deemed necessary for the commercialisation of an innovation [54].

5. Conclusion and implications for future research

This paper explores the applicability and implications of open innovation in agile systems development. In terms of future research it raises a number of significant questions that could be addressed. Some of the fundamental key ones could simply seek to examine the current state of innovative collaboration between the agile team and other stakeholders outside that business unit. This analysis should go well beyond the well-established on-site customer practice that usually involves one representative, often with a very focused and myopic role within that client organization. Researchers could examine the extent to which various intra- and inter-organisational stakeholders are typically involved in the agile development process, and how they contribute to the

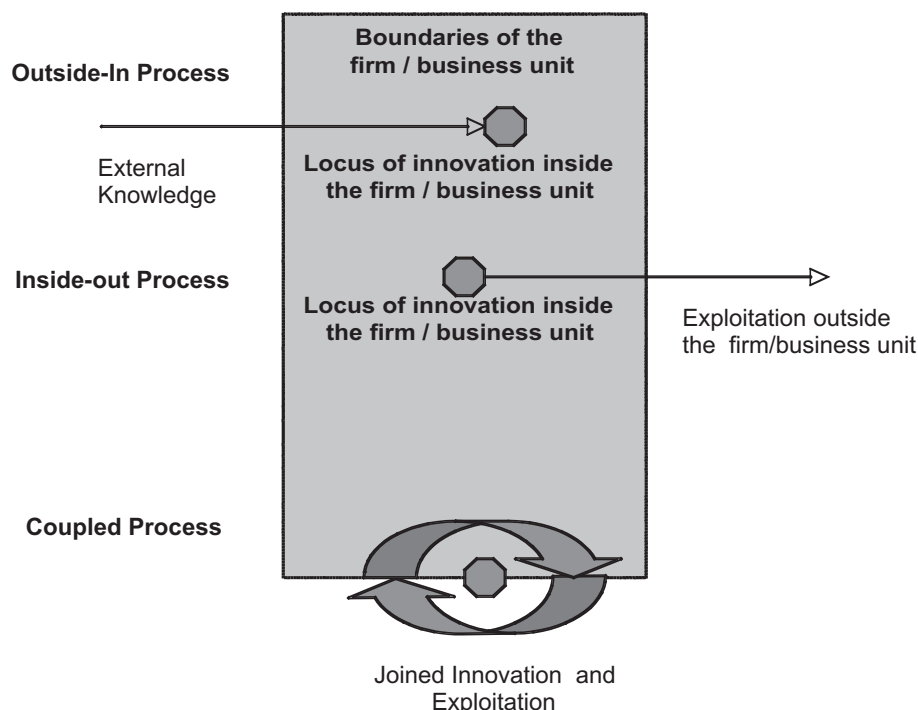


Fig. 1. Adapted framework for open innovation.

innovation process on these projects. Future research could also investigate the current barriers to open innovation that exist in agile development and what steps 'best practice' teams are using, if any, to overcome these barriers.

Additionally, research could also examine the role of the project manager in the transition towards open innovation. As we already know, in an agile development environment, the project manager's role is greatly changed, and is more akin to that of a facilitator or coordinator [3,43,46]. In agile projects, the organisation or team structure is "organic and flexible", as opposed to traditional structures which are "mechanistic, bureaucratic and formalized" [46]; the method is there not as a prescription, but something to be continuously tailored and moulded by the team [18,31]; the project is completed through a series of iterations, each often as short as a few working days [31,33], resulting in more frequent, short-term development; budgeting is more fluid and short-term [21], and software is valued over documentation [33]. Significantly, the customer plays a more continuous and embedded role, and thus is intrinsically involved in most project management decisions [6,7,29,36]. Moreover, developers are not confined to a specific specialised role and are encouraged to self-organise, interchanging and blending roles [46] and become involved in project management issues that may fall outside their traditional skill areas.

However, the very concept of incorporating open innovation principles may prove challenging for an agile project manager. As well as coping with managing in such a fluid, short-term environment, dealing with multiple projects and external entities adds further challenges and risk of unexpected outcomes. Indeed, identifying and engaging stakeholders in an intra and inter-organisational networks is one such challenge for the project manager. Introducing open innovation practices in an organisation implies change and is likely to result in uncertainty, risk and a great need for better coordination, which in turn may increase the current workload of a project manager. Nevertheless, mechanisms for scanning the project landscape should be incorporated into project management practices in agile organisations and project managers need to be aware that an IS project is no longer a local matter that can be treated as a closed innovation isolated from the rest of the organisation. After all, it has been found that project boundaries are pliable and negotiable and a project should be seen in light of other projects within an organisation. Indeed, knowledge in and about projects should be exchanged and individual projects should scan the open space of the organisation for other projects that constitute potential collaboration [28]. Thus, taking an open innovation route should present many additional benefits for an agile environment.

The conceptual framework proposed earlier also raises some interesting research questions. In terms of 'outside-in' open innovation, how should teams choose which suppliers, customers and third parties to collaborate with? This has not been the focus of any agile development research as far as we are aware. More importantly from an agile perspective given the tight and continuous interaction between all involved, how should they choose people to represent each party, and what are the characteristics of these representatives? Furthermore in an outside-in process, integration of the external knowledge gained is vital, but this can be quite challenging in an agile environment where knowledge transfer is almost exclusively tacit. Indeed, knowledge creation and generation, applying innovation across the organization and customer and supplier integration can be difficult to achieve in such an environment. Additionally, future research should seek to identify the organizational dimensions of absorptive capacity that are required in an agile environment to benefit from outside-in innovations.

The 'inside-out' open innovation process also throws up some relevant questions. Deciding to change the locus of knowledge-sharing by transferring ideas to stakeholders outside the business

unit or firm may prove challenging. Again this is exacerbated in an agile environment where explicit documentation and transfer of knowledge is usually greatly reduced. Similar to the outside-in approach, a successful inside-out approach may be contingent on the team's knowledge transfer capabilities and selection of appropriate stakeholders and their willingness and ability to engage and cooperate with the team. Additionally, we found it difficult to provide an illustrative case that combines both the inside-out and outside-in approach, i.e., the coupled process. Thus, how teams develop complementary internal value networks to create and gain external knowledge and ideas is a significant question. Where the coupled approach is enabled by a internal/external ecosystem or value network, there are more questions as to how the network is coordinated and maintained. Additionally, it is crucial to understand how governance is shared across the internal and external networks and how conflict is managed if it arises. Again, agile development is particularly difficult in this case where the traditional, formal, bureaucratic role of the project manager is diminished to that of a coach. This may work well on a small co-located team within a business unit, but may be very problematic when scaled up.

Other research areas beyond the conceptual framework include the question of how an organization can successfully manage the transition to a more open form of agile innovation is one that will be addressed. In addition, who will drive and who will resist the implementation of open innovation practices in an agile environment is another interesting arena. As the agile philosophy focuses on people and their creativity, we expect that individual attitude and acceptance will almost certainly be a prerequisite for the successful implementation of open innovation practices by the firm in question. Additionally, one could examine the capabilities and requirements needed for open innovation practices to be successful. Thus, further research in this area would be beneficial in providing an insight for researchers and practitioners alike, particularly focusing on the key benefits, challenges and best practices of open innovation in an agile environment.

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